Stem Cell Biology

How to Culture Pluripotent Stem Cells in Suspension

What are Stem Cells?

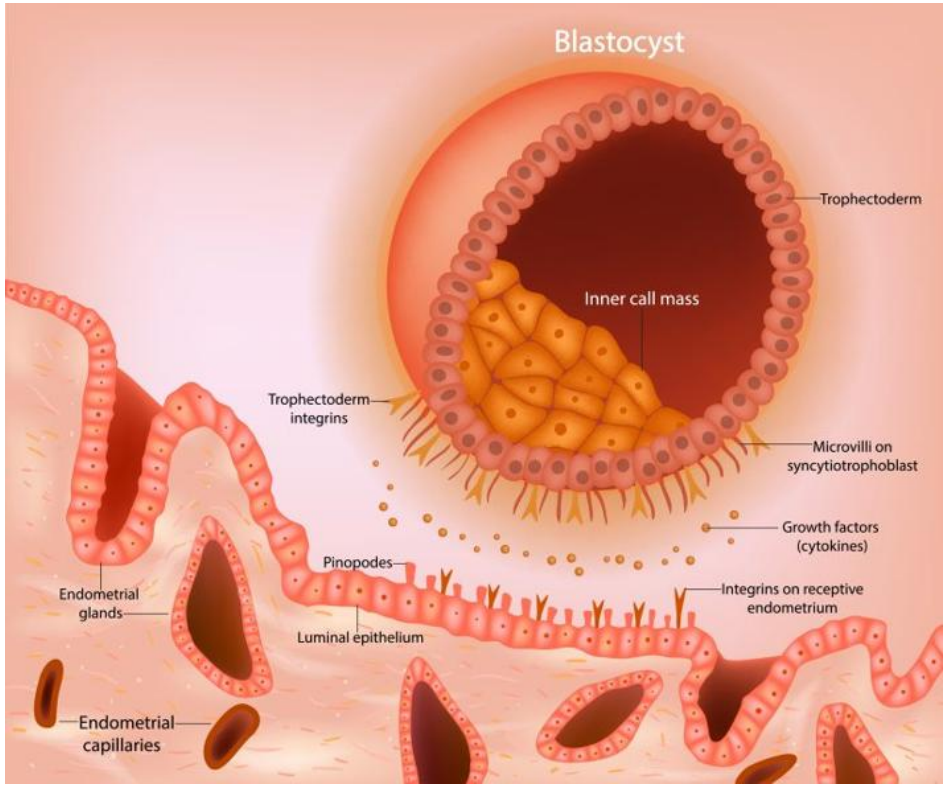
- **Stem cells** are undifferentiated cells capable of both self-renewal and differentiation.
  They can either remain in an undifferentiated state or specialize into specific cell types.
- **Two major categories exist**: adult (somatic) stem cells and embryonic stem cells.
  Embryonic stem cells are more flexible, as they can form any cell lineage.e **hESCs** represent the earliest and most potent form of stem cells in humans.

Pluripotency

- refers to the ability of a single cell to differentiate into derivatives of all three germ layers: ectoderm, mesoderm, and endoderm.
- This property allows hESCs to form neurons, muscles, hepatocytes, blood cells, and more.
- Pluripotency is regulated by specific transcription factors such as OCT4, SOX2, and NANOG.
- These genes maintain the undifferentiated state and repress differentiation pathways.
- When culture conditions change, these transcription factors are downregulated, allowing differentiation to begin.

Origin of hESCs

- Human embryonic stem cells are derived from the inner cell mass (ICM) of the blastocyst, a 5–7-day-old pre-implantation embryo.



Origin of hESCs

- At this stage, the blastocyst consists of an outer trophoblast and an inner cluster of cells—the ICM—which forms the embryo proper.
- Cells are isolated from the ICM using enzymatic or mechanical dissection.
- These isolated cells are then cultured on supportive feeder layers to establish stable cell lines.
- The resulting colonies of hESCs can proliferate indefinitely under the right conditions.

Sources of Embryos

- Most embryos used for hESC derivation come from IVF clinics where surplus embryos are no longer needed for reproduction.
- Some lines have also been derived from abnormal or aneuploid embryos that cannot result in viable pregnancies.

How to Culture Pluripotent Stem Cells in Suspension:

Laboratory Protocol: The 3D Suspension Strategy



Why Suspension?

Transitioning pluripotent stem cells (PSCs) from adherent plates into 3D suspension culture allows for massive scale-up, indefinite proliferation, and high-yield spheroid growth.

The Goal

To continually expand the cell population while strictly preventing inner core necrosis and unintended differentiation.

The Environment

Constant motion using CO_2-resistant orbital shakers and meticulously engineered chemical media.

Gemini Notebook

1- Initiation of PSC Suspension Culture

- **Purpose**: Transition adherent PSCs into suspension culture. Main Steps:
- Ensure aseptic conditions and wear PPE.
- Clean hood and surface with 70% ethanol.
- Prepare suspension culture vessels with complete StemScale PSC Medium + 10 µM Y-27632.
- Prewarm vessels at 37°C, 5% CO_2.

Materials:

- Gibco Complete StemScale Medium
- Gibco DPBS (no Ca^{2+}, no Mg^{2+})
- Gibco StemPro Accutase
- CO_2-resistant shaker


Gibco StemPro
Accutase cell
dissociation reagent
Gibco DPBS,
no calcium,
no magnesium
Gibco Complete StemScale Medium
Y-27632
compound
CO_2-resistant shaker
gibco
1X
DPBS
gibco
DPBS (1X)
Dulbecco's Phosphate Buffered Saline
[–] Calcium Chloride
[–] Magnesium Chloride
REF 14190-144 500 mL
LOT 2120423
2023-03-30
gibco
StemScale™
PSC Suspension Basal Medium
REF A48630-01 900 mL
LOT 1234567
1234-12-12
gibco
NUNC

Materials:

Procedure:

- **Aspirate** PSC medium from adherent cultures.
- **Wash** with DPBS and aspirate.
- Add **Accutase**, incubate 4–5 min at 37°C.
- **Triturate** gently → single-cell suspension.
- **Neutralize** with 3 mL StemScale Medium per 1 mL Accutase.
- **Centrifuge** → resuspend pellet in StemScale + Y-27632.
- **Count** viable cells and seed 100–150K cells/mL.
- Place on CO_2-resistant orbital **shaker** at:
- 70 RPM for 6-well plates
- 125 mL for shaker flasks



We recommend a speed of 70 RPM for 6-well plates and 125 mL shaker flask vessels.

2. Maintenance of PSC Suspension Culture

- **Purpose**: Support the growth and viability of PSC spheroids after initiation.
- **Observations**:
- PSC spheroids can be observed ~24 hours after initiation.
- **Feeding** Strategy:
- Use 50% medium replacement every other day.
- **Do not add Y-27632** compound during maintenance.
- We will need:
- Complete StemScale PSC Suspension Medium
- CO_2-resistant shaker
- P1000 pipette

Phase 2: Maintenance & Feeding Spheroids



The Gravity Settling Technique: Swirl the plate clockwise to gather spheroids, then tilt to 45° for 5 minutes. This allows spheroids to settle purely by gravity, avoiding the sheer damage of centrifugation.

- **Observation:** Spheroids become visible ~24 hours post-initiation.
- **Feeding:** Remove 50% of the old medium from the top. Add fresh complete StemScale PSC Suspension Medium. Return to shaker.

WARNING: Do not add Y-27632 during the maintenance phase.

Feed every other day (Day 1 through Day 4).

Gemini Notebook

Procedure:

- Swirl the culture plate **slowly clockwise** to gather spheroids at the center.
- Tilt the plate to **45°** and allow spheroids to **settle by gravity (≈5 min)**.
- **Remove 50%** of the old medium **from the top** (avoid aspirating spheroids).
- Add **fresh StemScale PSC Suspension Medium** (without Y-27632).
- Return to **37°C, 5% CO_2 incubator** on the shaker.
- **Notes:**
- Feeding is done **every other day (Day 1 → Day 4)**.
- Continue this process until spheroids reach the size suitable for passaging

3. Passaging of PSC Suspension Cultures

- **Purpose:**

  To prevent necrosis and maintain pluripotency by subculturing spheroids when they reach ~**400 µm** in diameter.

  This usually occurs **after 4–5 days of growth** in StemScale PSC Suspension Medium.

- **We will need:**
- Complete **StemScale PSC Medium**
- **StemPro Accutase** (cell dissociation reagent)
- **CO_2-resistant shaker**
- **Y-27632** compound
- **50 mL conical tubes**

Procedure:

- **Prepare** new suspension culture vessels as described in the initiation phase.
- **Swirl** the existing culture gently in a circular motion to gather spheroids in the center.
- **Collect** the spheroids by pipetting or pouring into a **50 mL conical tube**.
- **Wash** the vessel walls with fresh StemScale Medium to collect any remaining spheroids.
- **Centrifuge** the spheroids at **200 × g for 4 min**, then aspirate the spent medium.
- **Add** pre-warmed **StemPro Accutase** to the spheroid pellet.
- *Do not triturate using a P1000 pipette at this stage to avoid damaging the cells.*

Procedure:

- **Incubate** at **37°C for 10–15 minutes**, occasionally flicking or gently shaking to help dissociation.
- Once the suspension turns cloudy (indicating single-cell formation):
- **Pipette** up and down **5–7 times** to fully dissociate spheroids.
- **Neutralize** Accutase by adding **3 mL StemScale Medium per 1 mL Accutase**, and mix gently.
- **Centrifuge** again, then **resuspend** the pellet in fresh **StemScale PSC Suspension Medium + 10 µM Y-27632**.
- **Count** viable cells and **seed** at **100–150K cells/mL** into new non-treated suspension vessels.
- **Place** the vessels on the **CO_2-resistant orbital shaker** inside the incubator.